

Climate change hazards and Paths to sustainable future

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Abstract

Climate change impacts and responses are presently observed in physical and ecological systems. It is essential to understand how future climate change is going to affect our life. The purpose of this presentation to demonstrate that natural and human influences have caused the climate to change and how we can ensure a sustainable future.

Objective

BUSINESS AS USUAL

global temperatures : +3.2 C
air quality : 4.85 BILLION
lands protected : 1.4 %

SUSTAINABILITY PATH

global temperatures : + 1.6C
air quality : 0.66 BILLION
lands protected : 17%

Global climate change

EXTREME EVENTS & VARIABILITY

-FLOODS
-DROUGHT
-STORMS

IMPACT ON SECTORS

-SECURITY
-ECOSYSTEM

GROWTH IMPACTS

-PRODUCTIVITY
-CAPITAL
-SAFETY

DIRECT IMPACT

-ASSETS
-DESTRUCTION
-LIVES LOST

SLOW ONSET HAZARDS

-TEMPERATURE
-PRECIPITATION

TACKLING CLIMATE CHANGE

EDF's action to cut global warming pollution

OUR GOAL

Implement innovative strategies to reduce and eliminate greenhouse gas emissions and reduce the environmental consequences of climate change.

KEY

- PROBLEM
- EDF SOLUTION
- ENERGY
- WATER
- WASTE
- COMMUNITY ENGAGEMENT PROGRAMS
- GOVERNANCE
- CLIMATE ACTION PLAN

CLEAN-UP POWER PLANTS

Coal-fired power plants account for 40% of the nation's global warming pollution and emit a wide range of pollutants, including mercury, arsenic and selenium.

EDF is playing a key role in supporting the cleanup of coal-fired power plants. These toxic emissions threaten public health, harm thousands of birds and other species, and damage ecosystems.

MODERNIZE THE ELECTRIC GRID

American electric grid is outdated, incapable of managing the variable loads from renewable energy sources, and inefficient. EDF is helping power companies modernize their grids, making them more efficient and reliable. EDF is also helping power companies modernize their grids, making them more efficient and reliable.

MAKE BUILDINGS MORE ENERGY EFFICIENT

Businesses spend \$100 billion annually on energy to heat and cool commercial buildings, much of it lost due to poor insulation and outdated energy systems.

EDF is working with leading companies like GE, Facebook and Microsoft to reduce energy consumption and reduce their carbon footprint. EDF is also helping companies modernize their buildings, making them more efficient and reliable.

IMPROVE FUEL ECONOMY

The average fuel economy of cars in the U.S. is 24 mpg. EDF is working with leading companies like GM, Ford and Toyota to improve fuel economy and reduce greenhouse gas emissions.

EDF is also helping companies modernize their fleets, making them more efficient and reliable.

OTHER KEY INITIATIVES

DEFENDING CLEAN AIR

More than half of Americans live in areas that fail to meet federal health-based air standards. EDF is working with leading companies like GE, Facebook and Microsoft to reduce air pollution and improve air quality.

CALIFORNIA - THE PROVING GROUND

EDF is working with leading companies like GE, Facebook and Microsoft to reduce greenhouse gas emissions and improve air quality in California.

GLOBAL ACTION

EDF is helping companies modernize their operations, making them more efficient and reliable.

In China, the world's largest emitter, EDF is helping the government to modernize its power plants and reduce greenhouse gas emissions.

In India, EDF is helping the government to modernize its power plants and reduce greenhouse gas emissions.

In Vietnam, EDF is helping the government to modernize its power plants and reduce greenhouse gas emissions.

What can be done?



Conclusion

Adaptation can significantly reduce adverse impacts of climate change. It's an important part of social response to global climate change. Planned adaptation has the potential to reduce vulnerability & realize opportunities associated with climate change & hazards.

Reference : Environmental Defense Fund (EDF) , wiki and Institute for Risk and Disaster Reduction (IRDR) official webpage :